

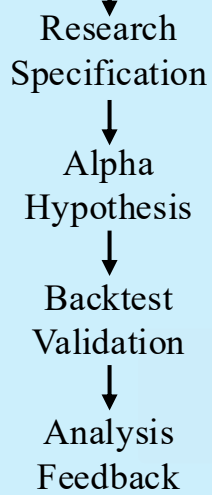
Prior Work

Open-loop Multi-Agent Trading System

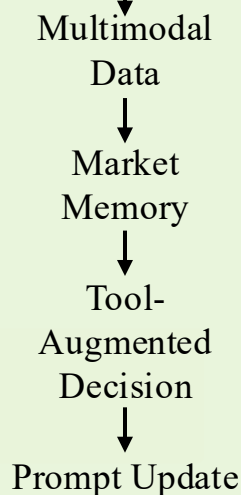
(a) Role Specialized



(b) Alpha Discovery



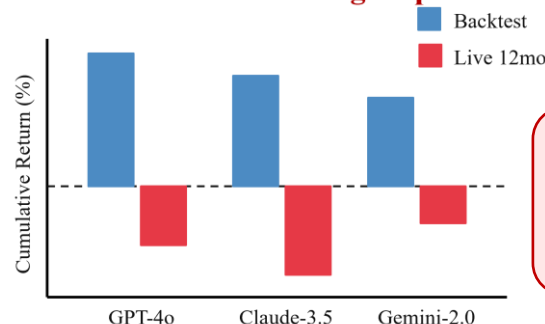
(c) Multimodal Tool-Augmented



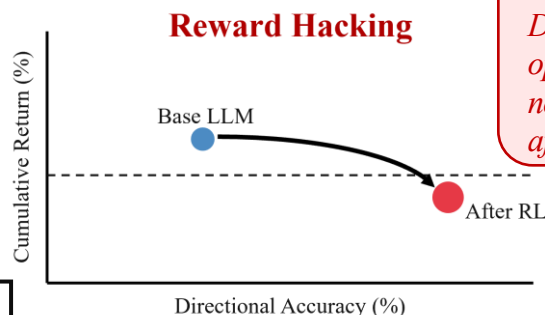
Their realized outcomes are **not** converted into risk-adjusted gradients on the policy parameters.

Observed Failure Modes

The Live Trading Gap



Failure Mode 1
Backtest gains do not reliably survive live deployment.

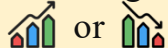


Failure Mode 2
Directional rewards optimize correctness, not portfolio utility after costs and risk.

FinOPD: On-Policy Distillation for Financial Self-Evolution

1. Hindsight PnL as Privileged Signal

Standard target o_t :



→ classification target

FinOPD teacher $o_t + \hat{y}$:

$\hat{y} = \{Sharpe, MDD, CVaR, Sortino\}$

→ risk-adjusted target

2. Marginal Agent Credit



5 agents weighted by contribution

3. Dual-Track Evolution

Parametric track

$$JSD(p_t || p_s) \rightarrow LoRA_{\theta}$$



Non-parametric track

$$(g, F, a, J) \rightarrow \text{episodic memory } M$$

4. Regime-gated Abstention



Trend / Volatile



Normal / Calm



Under a strict 17-month post-cutoff evaluation,
FinOPD achieves the strongest portfolio-level performance